

4-cycle tilings in a digraph

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Abstract

For a positive integer t , a t -cycle is a cycle of length t . A tiling of a graph G is a set of disjoint subgraphs (called tiles) contained in G . A tiling is a factor, if its union spans G . The main purpose of this paper is to explore the minimum semi-degree condition to guarantee a directed 4-cycle factor in digraphs. To do this, we first need a structural result which shows that every standard multigraph on $4k$ vertices with minimum degree $16k/3$ contains a tiling of size $k - 4$ such that any orientation of every tile has a directed 4-cycle. The degree condition is sharp up to an additive constant. Then using the absorbing method and the above result, we show that every digraph of sufficiently large order with minimum semi-degree at least $(2/3 + \varepsilon)n$ contains a directed 4-cycle factor, where $0 < \varepsilon < 1$ and $n \equiv 0 \pmod{4}$.

1 Introduction

In this paper all multigraphs considered are standard, that is, there exist at most two parallel edges between every pair of vertices; and all digraphs are simple, no parallel arcs exist but arcs with opposite directions are allowed. For terminology and notation not defined in this paper, we refer the readers to [5] and [11]. For a graph G and a vertex v in G , $d_G(v)$ denote the degree of v in G . The minimum degree of G is $\delta(G) = \min\{d_G(v) : v \in V(G)\}$. For a digraph D and a vertex v in D , $N_D^+(v)$ (resp., $N_D^-(v)$) and $d_D^+(v)$ (resp., $d_D^-(v)$) denote the out-neighborhood (resp., in-neighborhood) and out-degree (resp., in-degree) of v in D . The total degree or simply degree of v in D is $d_D(v) = d_D^-(v) + d_D^+(v)$. When it is clear from the context, we omit the subscript D or G . Like the undirected case, we can define the minimum out-degree $\delta^+(D)$, minimum in-degree $\delta^-(D)$ and minimum degree $\delta(D)$ of D , respectively. In particular, the minimum semi-degree of D is $\delta^0(D) = \min\{\delta^+(D), \delta^-(D)\}$. The *length* of a cycle or path C , denoted by $l(C)$, is the number of its edges. For a positive integer t , a t -cycle is a cycle of length t . A cycle or path in digraphs means a directed cycle or path. We use $T(t)$ (called t -triangle) and $Q(t)$ to denote a multigraph on three and four vertices with t edges, respectively. The multigraph $Q(9)$, which has five possible ways, is depicted in Figure 1.

A *tiling* of G is a set of disjoint subgraphs of G , called tiles. A tiling is a *factor*, if its union spans G . For a graph family $\mathcal{F}$, an $\mathcal{F}$ *tiling* is a tiling whose tiles are all isomorphic to a subgraph of $\mathcal{F}$. When $\mathcal{F}$ consists of a single graph F , an $\mathcal{F}$ tiling is just an F tiling. It is trivial to extend the above definitions to digraphs.

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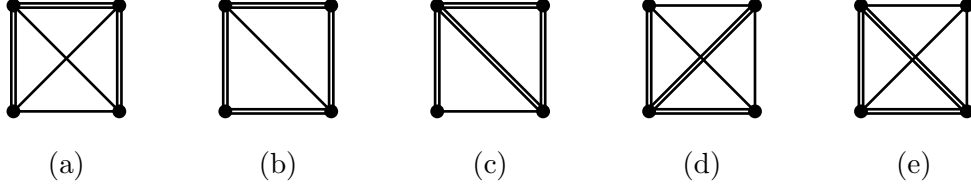


Figure 1: Possibilities for $Q(9)$. The family of multigraphs in (a)-(d) is denoted by $\mathcal{Q}^*(9)$.

A fundamental problem in graph theory is to find conditions for a graph to have an F factor. A classical result of Tutte [22] gives necessary and sufficient conditions for the case $F = K_2$. When F has a component consisting of at least 3 vertices the problem becomes more complicated, and simple characterizations like Tutte's theorem seem unlikely. Furthermore, Hell and Kirkpatrick [18] showed that for such a graph F , the decision problem of whether a graph has an F factor is NP-complete. The researchers therefore turn to find sufficient minimum degree conditions. Most notable is the celebrated Hajnal-Szemerédi Theorem [14] which states that every graph of order n with minimum degree $(1 - 1/r)n$ has a K_r factor, where $n \equiv 0 \pmod{r}$. The case $r = 3$ was proved earlier by Corrádi and Hajnal [8]. Over the last twenty years the Hajnal-Szemerédi Theorem has been generalised in a number of directions, see [2, 3, 10, 12, 15, 16, 17, 19, 20].

So given a digraph F , it is natural to seek degree conditions for a digraph D to contain an F factor. For a digraph there is more than one notion of degree, we mainly focus on the minimum semi-degree. In 2015, Treglown gave the directed version of the Hajnal-Szemerédi Theorem [21]: For any tournament T on r vertices, every digraph of sufficiently large order n with minimum semi-degree $(1 - 1/r)n$ contains a T factor, where r divides n . Recently, Cheng, Wang and Yan [7] gave a result on spanning F -subdivision of a digraph under the minimum semi-degree condition, for general F . As cycles are very basic but important structures, it is worthwhile to explore the minimum semi-degree condition for a digraph to contain a cycle factor. Particularly, the main result in [7] implies that for any real $0 < \alpha, \beta \ll 1$ there is a constant C_0 such that every digraph D with $\delta^0(D) \geq \frac{\lceil n_1 \rceil + \lceil n_2 \rceil + \dots + \lceil n_m \rceil}{2}$ contains m disjoint directed cycles of length $n_1, n_2, \dots, n_m$, for any integer partition $|V(D)| = n_1 + n_2 + \dots + n_m \geq C_0 m$ with the sum of the n_i less than αn being not more than βn . The above result shows that a digraph has cycle factors of any type, when its semi-degree is large and most of the cycles in the factor are not very short. It also indicates that the shorter the cycle lengths are in the cycle factor, the more complicated the problem is to find a cycle factor.

In fact, to find short cycle factors is an important topic in graphs. In 1963, Corrádi and Hajnal [8] proved that every graph G with $|G|$ divisible by 3 and $\delta(G) \geq 2|G|/3$ has a 3-cycle factor. For 4-cycle factors, there is a famous conjecture proposed by Erdős and Faudree [2]: If a graph G has $4n$ vertices with minimum degree at least $2n$, then G contains a 4-cycle factor. This conjecture has been solved in [24]. Therefore, it is meaningful to consider short cycle factor problems in digraphs.

Earlier before [7], Czygrinow, Kierstead, Molla [9] considered the minimum semi-degree condition for a digraph D to contain a directed 3-cycle factor and showed that for any real ε with $0 < \varepsilon < 1$, there exists $n_0 = n_0(\varepsilon)$ such that every digraph D on $n \geq n_0$ vertices with minimum semi-degree $\delta^0(D) \geq (2/3 + \varepsilon)n$ contains a directed 3-cycle factor, where $n \equiv 0 \pmod{3}$. Instead of proving the result directly in digraphs, they solved the problem by considering $T(5)$ tilings in multigraphs, which provides a new way to deal with problems in very dense digraphs. More precisely, they first showed

that every multigraph with minimum degree $(4n - 3)/3$ has a tiling of size $\lfloor n/3 \rfloor$, where one tile is a 4-triangle and the remaining $\lfloor n/3 \rfloor - 1$ tiles are 5-triangles. This result works as a foundational step for further proving, then using it and the absorbing method they got the result for directed 3-cycle factors in a digraph D under minimum semi-degree $(2/3 + \varepsilon)n$. Note that any orientation of a 5-triangle contains both a transitive 3-cycle and a directed 3-cycle, and any orientation of a 4-triangle contains a transitive 3-cycle. Therefore, the foundational result already implies that every digraph with $\delta(D) \geq (4n - 3)/3$ has an almost directed 3-cycle factor with only one exception. However, the minimum degree condition for a digraph to contain a directed 3-cycle factor is $(3n - 3)/2$, which is proved by Wang [23].

Motivated by [9] and followed their lines, in this paper we study the minimum semi-degree condition ensuring a digraph to contain a directed 4-cycle factor. Let $\mathcal{Q}^*(9)$ be the family of multigraphs shown in Figure 1 (a)-(d) and let Q_1 and Q_2 be the multigraphs depicted in Figure 2. Observe that for each multigraph in $\mathcal{Q}^*(9)$, it is isomorphic to Q_1 or it contains Q_2 as a subgraph. The following are our main results.

Theorem 1.1. *Let k be a positive integer. Every multigraph M on $4k$ vertices with minimum degree $\delta(M) \geq 16k/3$ contains a $\{Q_1, Q_2\}$ tiling of size $k - 4$.*

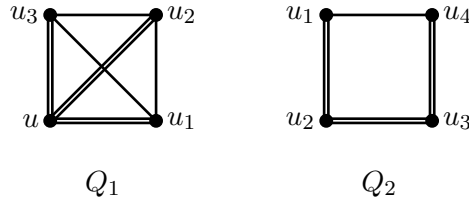


Figure 2: Multigraphs Q_1 and Q_2 .

All possible orientations of an undirected 4-cycle are depicted in Figure 3. Note that no matter how we orient the edges of Q_1 and Q_2 , both C_4 , F_3 and F_2 can be obtained. So Theorem 1.1 implies immediately the following corollary.

Corollary 1. *Let k be a positive integer. For all integers $a, b, c \geq 0$ with $a + b + c = k - 4$, every digraph D on $4k$ vertices with minimum degree $\delta(D) \geq 16k/3$ contains a tiling consisting of a copies of C_4 , b copies of F_3 and c copies of F_2 , where C_4 , F_3 and F_2 are depicted in Figure 3.*

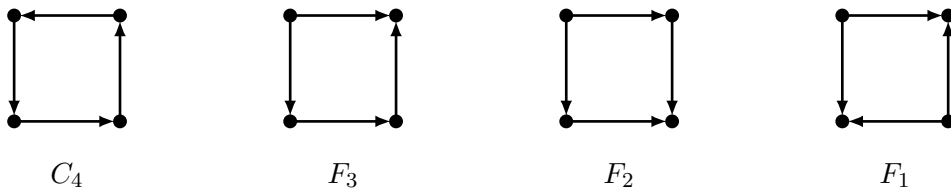


Figure 3: All possible orientations of an undirected 4-cycle. Note that the length of the longest directed path in F_i is i .

Using Theorem 1.1 and the absorbing method, we get the following semi-degree condition for a digraph to contain a directed 4-cycle factor.

Theorem 1.2. *For any real number ε with $0 < \varepsilon < 1$, there exists $n_0 = n_0(\varepsilon)$ such that every digraph D on $n \geq n_0$ vertices with minimum semi-degree $\delta^0(D) \geq (2/3 + \varepsilon)n$ contains a 4-cycle factor, where $n \equiv 0 \pmod{4}$.*

The proofs of Theorems 1.1 and 1.2 are presented in Sections 2 and 3, respectively. In Section 4, we show that the degree condition in Theorem 1.1 is sharp up to an additive constant and mention a related problem about all possible cycle factors in a digraph.

To end this section, we prepare terminology and notation, which will be used in the proofs. Given a (multi)graph $G = (V, E)$, for two vertices $x, y \in V$, let $e_G(x, y)$ be the number of edges with ends x and y . An edge xy in G is heavy if $e_G(x, y) = 2$ and it is light if $e_G(x, y) = 1$. If xy is heavy (or light), we say x is heavy (or light) adjacent to y . Let $N_G(v)$ be the neighborhood of a vertex v in G . For a subset $X \subseteq V$, we use $G[X]$ to denote the subgraph induced by X in G and let $G - X = G[V \setminus X]$. For two disjoint subgraphs F_1 and F_2 of G , let $e_G(V(F_1), V(F_2))$ denote the number of edges having one end in $V(F_1)$ and the other in $V(F_2)$. We often identify a subgraph F of G with its vertex set $V(F)$. For example, we write $G - F$ and $e_G(F_1, F_2)$ instead of $G - V(F)$ and $e_G(V(F_1), V(F_2))$. We often denote a bipartite graph B with partite sets V_1, V_2 by $B(V_1, V_2)$. The complete bipartite graph with one partite of size s and the other of size t is denoted by $K_{s,t}$. Denote by $K_{s,t}^-$ the graph obtained from $K_{s,t}$ by deleting exactly one edge. For a positive integer t , let $[t]$ denote the set $\{1, 2, \dots, t\}$.

2 Proof of Theorem 1.1

In this section, let $M = (V, E)$ be a multigraph. A subgraph in M is heavy (or light) if its every edge is heavy (or light). The set of end vertices of a path P is denoted by $\text{End}(P)$. The multigraphs Q_1 and Q_2 which depicted in Figure 2 are called *feasible* multigraphs. A multigraph is said to be *r-feasible* if it contains r disjoint feasible multigraphs.

Now we give a sketch of the proof of Theorem 1.1. Let M be a multigraph satisfying the condition of Theorem 1.1. Partition M into two parts, one part (part one) is a collection of maximum number of disjoint feasible graphs, the other part is the remaining graph. If we can get three disjoint heavy paths P_1, P_2, P_3 of length 2 in the remaining graph, then the degree condition implies that the end vertices of P_1, P_2, P_3 have many neighbors in some feasible graph Q^* in part one. In the subgraph induced by Q^*, P_1, P_2 and P_3 , we can easily find two disjoint feasible graphs, which increases the number of disjoint feasible graphs and gives a contradiction. We show that every three disjoint heavy edges in the remaining graph produce a heavy path of length 2. All we need is to find enough disjoint heavy edges in the remaining graph, which is the most technical part. We show that every six vertices in the remaining graph produce a heavy edge. Since we are not trying to find a factor but only an almost factor in this section, there are enough vertices in the remaining graph for us to use.

2.1 Preparations for Theorem 1.1

In this section, the graph Q_1 every time we mention is the feasible graph depicted in Figure 2. Its four vertices are u and $u_i, i \in [3]$, where u is heavy adjacent to u_i for each $i \in [3]$ and u_1, u_2, u_3 form a light triangle. We call u the *center* vertex of Q_1 . And the graph Q_2 is also always the feasible graph depicted in Figure 2, where its four vertices are u_1, u_2, u_3, u_4 , and $u_1 u_2 u_3 u_4$ is a heavy path of length 3 and $u_1 u_4$ is a light edge.

The following facts are useful and are not very hard to check, so we omit the proofs.

Fact 1. Let Q be a feasible graph and v a vertex outside Q . If $e_M(v, Q) \geq 4 + i$, then there exist at least i heavy edges between v and Q .

Fact 2. Let $T(5)$ be a 5-triangle and v be a vertex outside $T(5)$. If $e_M(v, T(5)) \geq 4$, then $M[V(T(5)) \cup \{v\}]$ contains a feasible graph Q .

Fact 3. (i) Let P and P' be two disjoint heavy paths of length at most two. If $e_M(\text{End}(P), \text{End}(P')) \geq 5$, then $M[V(P \cup P')]$ contains a feasible graph Q_2 .

(ii) Let $U, U' \subseteq V$ be two disjoint vertex sets with $|U| = |U'| = 2$. If $e_M(U, U') \geq 7$, then $M[U \cup U']$ contains a feasible graph Q_2 .

The following Lemmas 1 to 3 are used to find a heavy edge in the remaining graph.

Lemma 1. Let Q_1 be the feasible graph and W a vertex set of size 6 outside Q_1 . If $e_M(W, Q_1) \geq 33$, then $M[W \cup V(Q_1)]$ contains a feasible graph and a heavy edge, which are disjoint.

Proof. Suppose that there exists a vertex $v_1 \in W$ with $d_{Q_1}(v_1) = 8$, i.e. v_1 is heavy adjacent to every vertex of Q_1 . Since $e_M(W, Q_1) \geq 33$, there is another vertex $v_2 \in W$ such that $d_{Q_1}(v_2) \geq 5$. By Fact 1, v_2 is heavy adjacent to at least one vertex of Q_1 , say v . Then the subgraph induced by $(V(Q_1) \setminus \{v\}) \cup \{v_1\}$ contains a feasible graph and v_2v is a heavy edge disjoint from it.

So assume that every vertex of W has degree at most 7 in Q_1 . Then $e_M(W, Q_1) \geq 33$ implies that there are at least two vertices $v_1, v_2 \in W$ such that $d_{Q_1}(v_1), d_{Q_1}(v_2) \geq 6$. By Fact 1, v_2 is heavy adjacent to at least two vertices of Q_1 , and at least one of them is u_i for some i , say u_3 . Since u, u_1, u_2 form a $T(5)$ and $d_{Q_1-u_3}(v_1) \geq d_{Q_1}(v_1) - 2 = 4$, Fact 2 implies that the subgraph induced by v_1, u, u_1, u_2 contains a feasible graph, and v_2u_3 is a heavy edge disjoint from it. $\square$

We define $(d_1, d_2, \dots, d_n)$ with $d_1 \geq d_2 \geq \dots \geq d_n$ to be a degree sequence from X to Y if there exist n vertices $v_1, v_2, \dots, v_n$ in X such that $d_Y(v_i) \geq d_i$ for each $1 \leq i \leq n$. To prove Lemmas 2 and 3, we need the following observations.

Observation 1. Let Q_2 be the feasible graph and W a vertex set outside Q_2 . If $(8, 5)$ or $(7, 6)$ is a degree sequence from W to Q_2 , then $M[W \cup V(Q_2)]$ contains a feasible graph and a heavy edge, which are disjoint.

Proof. **Case A.** $(8, 5)$ is a degree sequence from W to Q_2 . Let $v_1, v_2 \in W$ such that $d_{Q_2}(v_1) = 8$ and $d_{Q_2}(v_2) \geq 5$. By Fact 1, v_2 is heavy adjacent to at least one vertex u_i of Q_2 . Then $M[(V(Q_2) \setminus \{u_i\}) \cup \{v_1\}]$ contains a feasible graph which is disjoint with the heavy edge v_2u_i .

Case B. $(7, 6)$ is a degree sequence from W to Q_2 . Let $v_1, v_2 \in W$ such that $d_{Q_2}(v_1) \geq 7$ and $d_{Q_2}(v_2) \geq 6$. Since $d_{Q_2}(v_1) \geq 7$, both $e(v_1, \{u_1, u_3\}) \geq 3$ and $e(v_1, \{u_2, u_4\}) \geq 3$. Note that both $u_1u_2u_3$ and $u_2u_3u_4$ are heavy paths in Q_2 . Thus $e(v_1, \{u_1, u_3\}) \geq 3$ and $e(v_1, \{u_2, u_4\}) \geq 3$ implies that both $v_1u_1u_2u_3$ and $v_1u_2u_3u_4$ are feasible graphs containing Q_2 . By Fact 1 and $d_{Q_2}(v_2) \geq 6$, v_2 is heavy adjacent to at least 2 vertices of Q_2 . If v_2u_1 is a heavy edge, then $v_1u_2u_3u_4$ and v_2u_1 are the disjoint graphs desired. Similarly, if v_2u_4 is a heavy edge, then $v_1u_1u_2u_3$ and v_2u_4 are the disjoint graphs desired. Therefore, the edges between v_2 and Q_2 are fixed, v_2u_2, v_2u_3 are heavy edges and v_2u_1, v_2u_4 are light edges. Since $d_{Q_2}(v_1) \geq 7$, either v_1u_1 or v_1u_4 is a heavy edge. If v_1u_1 is heavy, then

$v_2u_2u_3u_4$ is a feasible graph isomorphic to Q_2 disjoint from it. Otherwise, v_1u_4 is heavy and $v_2u_1u_2u_3$ is a feasible graph isomorphic to Q_2 disjoint from it. $\square$

Observation 2. Let Q_2 be the feasible graph and W a vertex set of size at least 2 outside Q_2 . Suppose that every vertex in W has degree exactly 6 in Q_2 . If $M[W \cup V(Q_2)]$ does not contain a feasible graph and a heavy edge which are disjoint, then in $M[W \cup V(Q_2)]$, W is an independent set and every vertex in W is heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 (type 1) or is heavy adjacent to u_1, u_3 and light adjacent to u_2, u_4 (type 2) or is heavy adjacent to u_2, u_4 and light adjacent to u_1, u_3 (type 3). The latter two cases are symmetric.

Proof. Suppose that there is some $v_1 \in W$ such that both $e(v_1, \{u_1, u_3\}) \geq 3$ and $e(v_1, \{u_2, u_4\}) \geq 3$. Since both $u_1u_2u_3$ and $u_2u_3u_4$ are heavy paths, both $v_1u_1u_2u_3$ and $v_1u_2u_3u_4$ are feasible graphs isomorphic to Q_2 . This means that no other vertex in W can be heavy adjacent to u_1 or u_4 , otherwise one may find a feasible graph and a heavy edge which are disjoint. Therefore, all the other vertices in W are heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 . However, in this case any other vertex $v_2 \in W$ and $v_2 \neq v_1$ also satisfies that both $e(v_2, \{u_1, u_3\}) \geq 3$ and $e(v_2, \{u_2, u_4\}) \geq 3$. By the above argument, v_1 can not be heavy adjacent to u_1 or u_4 . Hence, v_1 is also heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 . Now every vertex in W is heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 . Next we claim that W is an independent set in $M[W]$. If v_1v_2 is an edge in $M[W]$, then from the above structure, $M[\{u_3, v_1, v_2, u_4\}]$ is isomorphic to Q_1 , where $v_1v_2u_4$ is the light triangle and u_3 is heavy adjacent to every vertex on it. Note that u_1u_2 is heavy. Then u_1u_2 and $M[\{u_3, v_1, v_2, u_4\}]$ are the desired disjoint graphs.

So there is a vertex $v_1 \in W$ such that $e(v_1, \{u_1, u_3\}) = 4$, $e(v_1, \{u_2, u_4\}) = 2$ or $e(v_1, \{u_1, u_3\}) = 2$, $e(v_1, \{u_2, u_4\}) = 4$. We only prove the former case, the latter case can be proved similarly. In the case, $M[\{v_1, u_1, u_2, u_3\}]$ and $M[\{u_1, v_1, u_3, u_4\}]$ are both feasible graphs containing Q_2 . So no other vertex in W can be heavy adjacent to u_2 or u_4 , which indicates that all the other vertices in W are heavy adjacent to u_1, u_3 and light adjacent to u_2, u_4 . So every vertex $v_2 \neq v_1$ and $v_2 \in W$ satisfies that $e(v_1, \{u_1, u_3\}) = 4$, $e(v_1, \{u_2, u_4\}) = 2$, which implies that v_1 is also heavy adjacent to u_1, u_3 and light adjacent to u_2, u_4 . The structure above illustrates that W is independent in M . Otherwise, say $v_1v_2 \in M[W]$, then $M[\{v_1, v_2, u_3, u_4\}]$ is isomorphic to Q_1 , where $v_1v_2u_4$ is a light triangle and u_3 is heavy adjacent to all the vertices on it. Therefore, u_1u_2 and $M[\{v_1, v_2, u_3, u_4\}]$ are the desired disjoint graphs, which completes the proof. $\square$

Lemma 2. Let Q_2 be the feasible graph and W a vertex of size 6 outside Q_2 . If $e_M(W, Q_2) \geq 33$ and $M[W \cup V(Q_2)]$ does not contain a feasible graph and a heavy edge which are disjoint, then $M[W \cup V(Q_2)]$ contains a feasible graph and a light $K_{2,4}^-$ which are disjoint.

Proof. From Observation 1, there is no degree sequence $(8, 5)$ or $(7, 6)$ from W to Q_2 . So there are only three types of degree sequences from W to Q_2 , which are $(6, 6, 6, 5, 5, 5)$, $(6, 6, 6, 6, 5, 4)$ and $(6, 6, 6, 6, 6, 3)$. Moreover, every vertex in W whose degree is at least 6 in Q_2 is exactly 6, otherwise there is a $(7, 6)$ degree sequence. Let $W' = \{v \in W : e(v, Q_2) = 6\}$ and $W'' = W - W'$. In what follows, we prove that only using the edges between Q_2 and W is enough to find the structures desired. By Observation 2, the structures between W' and Q_2 and inside W' are fixed in three different types. We may first consider the case that the structure between W' and Q_2 is of type 1, that is, every vertex in W' is heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 .

In this case, any vertex in W'' is not heavy adjacent to u_1 or u_4 , otherwise as we discussed many times before, there exists a feasible graph and a heavy edge which are disjoint. If the degree sequence is $(6, 6, 6, 5, 5, 5)$, then $|W''| = 3$ and every vertex in W'' has degree 5 in Q_2 . Let $W' = \{v_1, v_2, v_3\}$ and $W'' = \{w_1, w_2, w_3\}$. If $e(w_i, \{u_2, u_3\}) \leq 3$ for each $1 \leq i \leq 3$, then w_i is light adjacent to both u_1 and u_4 , as $e(w_i, Q_2) = 5$. Then $B(\{u_1, u_4\}, \{W' \cup \{v_3\}\})$ contains a light $K_{2,4}^-$. Moreover, $v_1 u_2 v_2 u_3 v_1$ is a heavy cycle, which is obviously disjoint with the above $K_{2,4}^-$ and we are done. If there is only one vertex say $w_1 \in W''$ such that $e(w_1, \{u_2, u_3\}) \leq 3$, then w_2 and w_3 are heavy adjacent to both u_2, u_3 . Now the bipartite graph $B(\{u_1, u_4\}, \{W' \cup \{w_1\}\})$ contains a light $K_{2,4}^-$, and $w_2 u_2 w_3 u_3 w_2$ is a heavy cycle disjoint with the above $K_{2,4}^-$. So we may assume there are exactly two vertices $w_1, w_2 \in W''$ such that $e(w_1, \{u_2, u_3\}) \leq 3$ and $e(w_2, \{u_2, u_3\}) \leq 3$. Now $B(\{u_1, u_4\}, \{v_1, v_2, w_1, w_2\})$ and $v_3 u_2 w_3 u_3 v_3$ are what we want.

Next we consider the case when the degree sequence is $(6, 6, 6, 6, 5, 4)$. Let $W' = \{v_1, v_2, v_3, v_4\}$ and $W'' = \{w_1, w_2\}$. We may w.l.o.g assume that $e(w_1, Q_2) = 5$. If both $e(w_1, \{u_1, u_4\}) \leq 1$ and $e(w_2, \{u_1, u_4\}) \leq 1$, then $e(w_2, \{u_2, u_3\}) \geq 3$ and $e(w_1, \{u_2, u_3\}) = 4$. Now $u_2 w_1 u_3 w_2 u_2$ is a feasible graph Q and $B(\{u_1, u_4\}, W')$ contains a light $K_{2,4}^-$ which is disjoint with Q . So at most one vertex w_i in W'' satisfies $e(w_i, \{u_1, u_4\}) \leq 1$, then $e(w_i, \{u_2, u_3\}) \geq 3$ and $e(w_{3-i}, \{u_1, u_4\}) = 2$. Now $u_2 v_4 u_3 w_i w_2$ is a feasible graph Q and $B(\{u_1, u_4\}, \{v_1, v_2, v_3, w_{3-i}\})$ contains a $K_{2,4}^-$ which is disjoint with Q . Finally we consider the case when the degree sequence is $(6, 6, 6, 6, 6, 3)$. Let $W' = \{v_1, v_2, v_3, v_4, v_5\}$ and $W'' = \{w_1\}$. If $e(w_1, \{u_1, u_4\}) \geq 1$, then $G[\{u_2, v_4, u_3, v_5, u_2\}]$ is a feasible graph and there is a light $K_{2,4}^-$ in $B(\{u_1, u_4\}, \{v_1, v_2, v_3, w_1\})$. So we may assume that $e(w_1, \{u_2, u_3\}) \geq 3$. Now $G[\{u_2, v_5, u_3, w_1, u_2\}]$ is a feasible graph and $B(\{u_1, u_4\}, \{v_1, v_2, v_3, v_4\})$ contains a light $K_{2,4}^-$.

From the above analysis, we see that in any case of the three degree sequences, always use u_1, u_4 as one part to construct the graph $K_{2,4}^-$. In the case that the structure between W' and Q is of type 2 (resp., type 3), we also need to discuss three cases according to the three different degree sequences, but in this case always use u_2, u_4 (resp., u_1, u_3) as one part to construct the graph $K_{2,4}^-$. All the details are similar to the type 1 so we omit the proof. $\square$

Lemma 3. *Suppose that Q_2 and $K_{2,4}^-$ are disjoint and $e_M(K_{2,4}^-, Q_2) \geq 33$. Then $M[V(K_{2,4}^- \cup Q_2)]$ contains a feasible graph and a heavy edge, which are disjoint.*

Proof. We may assume that $K_{2,4}^-$ is light otherwise there is nothing to prove. Suppose that $V(K_{2,4}^-) = \{v_1, v_2, w_1, w_2, w_3, w_4\}$, where $\{v_1, v_2\}$ is one partite and $\{w_1, w_2, w_3, w_4\}$ is the other partite. If there is a degree sequence of $(8, 5)$ or $(7, 6)$ from $K_{2,4}^-$ to Q_2 , then by Observation 1 we are done. So the only possible degree sequences from $K_{2,4}^-$ to Q_2 are $(6, 6, 6, 5, 5, 5)$, $(6, 6, 6, 6, 5, 4)$ and $(6, 6, 6, 6, 6, 3)$. Let $W' = \{v \in K_{2,4}^- : e(v, Q_2) = 6\}$. By Observation 2, W' is an independent set in $M[V(K_{2,4}^-)]$, so the degree sequence $(6, 6, 6, 6, 6, 3)$ is impossible. In the other two cases $(6, 6, 6, 5, 5, 5)$ and $(6, 6, 6, 6, 5, 4)$, Observation 2 indicates $W' \subseteq \{w_1, w_2, w_3, w_4\}$, which means that at least three vertices of $\{w_1, w_2, w_3, w_4\}$ are of degree 6. Since $V(K_{2,4}^-)$ has at most one vertex of degree less than 5 and $K_{2,4}^- \subset M[V(K_{2,4}^-)]$, there is an edge $v_i w_j \in M[V(K_{2,4}^-)]$, where $e(v_i, Q_2) \geq 5$ and $e(w_j, Q_2) = 6$. W.l.o.g we may assume that $v_i w_j = v_1 w_1$. Observation 2 shows that w_1 is heavy adjacent to exactly two vertices of Q_2 . Note that type 2 and type 3 are symmetric, so the proof can be divided into the following to cases.

Case 1. w_1 is heavy adjacent to u_2, u_3 and light adjacent to u_1, u_4 .

If v_1 is heavy adjacent to u_1 or u_4 , then we are done. So v_1 is heavy adjacent to at least one of

u_2 and u_3 . If v_1 is heavy adjacent to both u_2, u_3 , then v_1 is light adjacent to one of u_1, u_4 , say u_1 . Then $M[\{u_2, u_1, v_1, w_1\}]$ contains a Q_1 where $u_1v_1w_1$ is a light triangle and u_2 is heavy adjacent to the other three vertices. Now u_3u_4 is a heavy edge disjoint with Q_1 . If v_1 is heavy adjacent to one of u_2, u_3 , say u_2 , then v_1 is light adjacent to both u_1, u_4 . Now still $M[\{u_2, u_1, v_1, w_1\}]$ contains a Q_1 where $u_1v_1w_1$ is a light triangle and u_2 is heavy adjacent to the other three vertices, and u_3u_4 is a heavy edge disjoint from it.

Since $e(v_1, Q_2) \geq 5$, either v_1 is heavy adjacent to u_1 or u_4 , or one of $\{u_1, u_2\}$ and $\{u_3, u_4\}$, say $\{u_1, u_2\}$, such that v_1 is heavy adjacent to u_2 and light adjacent to u_1 .

Case 2. w_1 is heavy adjacent to u_1, u_3 and light adjacent to u_2, u_4 .

If v_1 is heavy adjacent to u_2 or u_4 , then we are done. So v_1 is heavy adjacent to at least one of u_1, u_3 . If v_1 is heavy adjacent to both u_1, u_3 , then v_1 is light adjacent to one of u_2, u_4 , say u_2 . Then $M[\{u_1, u_2, v_1, w_1\}]$ contains a Q_1 where $u_2v_1w_1$ is a light triangle and u_1 is heavy adjacent to the other three vertices. Now u_3u_4 is a heavy edge disjoint with Q_1 . If v_1 is heavy adjacent to one of u_1, u_3 , say u_1 , then v_1 is light adjacent to both u_2, u_4 . Now still $M[\{u_1, u_2, v_1, w_1\}]$ contains a Q_1 where $u_1v_1w_1$ is a light triangle and u_2 is heavy adjacent to the other three vertices, and u_3u_4 is a heavy edge disjoint from it. $\square$

The purpose of the following Lemmas 4 and 5 is to find a heavy path of length 2 in the remaining graph.

Lemma 4. Let $W = \{v_1v_2, v_3v_4, v_5v_6\}$ be a heavy matching of size 3, which means each matching edge is a heavy edge. If Q_1 is the feasible graph which is disjoint from W and $e_M(W, Q_1) \geq 33$, then $M[W \cup V(Q_1)]$ contains a feasible graph Q and a heavy path of length 2, which are disjoint.

Proof. Since $e_M(W, Q_1) \geq 33$, one heavy edge of W , say v_1v_2 , satisfy $e_M(\{v_1, v_2\}, Q_1) \geq 11$. By average, one of v_1 and v_2 , say v_1 , has $e_M(v_1, Q_1) \geq 6$. On the other hand, $e_M(\{v_1, v_2\}, Q_1) \leq 16$, so $e_M(W - \{v_1, v_2\}, Q_1) \geq 17$, which implies one of $\{v_3, v_4, v_5, v_6\}$, say v_3 , has $e_M(v_3, Q_1) \geq 5$. By Fact 1, there is at least one heavy edge between v_3 and Q_1 . If v_3u_i is an heavy edge for some $i \in [3]$, then $u_iv_3v_4$ is a heavy path of length 2 and $Q_1 - u_i$ is a $T(5)$. Since $e_M(v_1, Q_1) \geq 6$ and $e_M(v_1, u_i) \leq 2$, we have $e_M(v_1, Q_1 - u_i) \geq 4$. Then by Fact 2, $M[(Q_1 - u_i) \cup \{v_1\}]$ contains a feasible graph Q . In this case, we finish the proof. Now we may assume that $e_M(v_3, u_i) \leq 1$, for each $i \in [3]$. By $e_M(v_1, Q_1) \geq 6$ and Fact 1, there are at least two heavy edges between v_1 and Q_1 . Obviously, v_1u_i is a heavy edge for some $i \in [3]$. Then $u_iv_1v_2$ is a heavy path of length 2 and $Q_1 - u_i$ is a $T(5)$. Since $e_M(v_3, Q_1) \geq 5$ and $e_M(v_3, u_i) \leq 1$, we have that $e_M(v_3, Q_1 - u_i) \geq 4$. Fact 2 again implies that $M[(Q_1 - u_i) \cup \{v_1\}]$ contains a feasible graph Q . Now in any case we finish the proof. $\square$

Lemma 5. Let $W = \{v_1v_2, v_3v_4, v_5v_6\}$ be a heavy matching of size 3, which means each matching edge is a heavy edge. If Q_2 is the feasible graph which is disjoint from W and $e_M(W, Q_2) \geq 33$, then $M[W \cup V(Q_2)]$ contains a feasible graph Q and a heavy path of length 2, which are disjoint.

Proof. If there is an edge say $v_1v_2 \in W$ such that $e(\{v_1, v_2\}, Q_2) \geq 13$. Then both $e(\{v_1, v_2\}, \{u_1, u_2\}) \geq 5$ and $e(\{v_1, v_2\}, \{u_3, u_4\}) \geq 5$. Then by Fact 3 (i), both $M[\{v_1, v_2, u_1, u_2\}]$ and $M[\{v_1, v_2, u_3, u_4\}]$ contain feasible graphs. Since $e_M(W, Q_2) \geq 33$, at least one of the other two edges, say $v_3v_4 \in W$, satisfies $e(\{v_3, v_4\}, Q_2) \geq 9$. By average, $e(\{v_3, v_4\}, \{u_1, u_2\}) \geq 5$ or $e(\{v_3, v_4\}, \{u_3, u_4\}) \geq 5$. Without

loss of generality, assume $e(\{v_3, v_4\}, \{u_3, u_4\}) \geq 5$. Now Fact 3 (i) implies that both $M[\{v_1, v_2, u_1, u_2\}]$ and $M[\{v_3, v_4, u_3, u_4\}]$ have feasible graphs, we are done.

So every edge in W has degree at most 12 in Q_2 . Since $e_M(W, Q_2) \geq 33$, at least two edges, say v_1v_2, v_3v_4 of W , satisfy $e(\{v_1, v_2\}, Q_2) \geq 11$ and $e(\{v_3, v_4\}, Q_2) \geq 11$. From the above argument we know that $e(\{v_1, v_2\}, \{u_1, u_2\}) \leq 4$ or $e(\{v_1, v_2\}, \{u_3, u_4\}) \leq 4$. Without loss of generality, assume $e(\{v_1, v_2\}, \{u_3, u_4\}) \leq 4$. Then $e(\{v_1, v_2\}, \{u_1, u_2\}) \geq 7$. If $e(\{v_3, v_4\}, \{u_2, u_3\}) \leq 4$, then $e(\{v_3, v_4\}, \{u_1, u_4\}) \geq 7$. By Fact 3 (ii), $M[\{v_3, v_4, u_1, u_4\}]$ contains a feasible graph Q . Since $e(\{v_1, v_2\}, \{u_1, u_2\}) \geq 7$, we have $e(\{v_1, v_2\}, u_2) \geq 3$, either $v_1v_2u_2$ or $u_2v_1v_2$ is a heavy path of length 2 which is disjoint from Q . Now consider the case when $e(\{v_3, v_4\}, \{u_2, u_3\}) \geq 5$. Then $M[\{v_3, v_4, u_2, u_3\}]$ contains a feasible graph Q . Similarly, $e(\{v_1, v_2\}, \{u_1, u_2\}) \geq 7$ indicates $e(\{v_1, v_2\}, u_1) \geq 3$, either $v_1v_2u_1$ or $u_1v_1v_2$ is a heavy path of length 2 which is disjoint from Q . $\square$

Lemmas 4 and 5 imply the following proposition, which makes heavy matchings into heavy paths of length 2. Heavy paths of length 2 are helpful when getting the desired structures.

Proposition 1. *Let $W = \{v_1v_2, v_3v_4, v_5v_6\}$ be a heavy matching of size 3, and Q a feasible graph. If W and Q are disjoint and $e_M(W, Q) \geq 33$, then $M[W \cup V(Q)]$ contains a feasible graph Q' and a heavy path of length 2, which are disjoint.*

The following is the key instrument to get a contradiction in our proof, which says that if there are many edges between a feasible graph and the end vertices of several disjoint heavy paths of length 2, then easily we may find another desired feasible graph.

Proposition 2. *Let P_1, P_2, P_3 be three disjoint heavy paths of length 2 and $W = \bigcup_{i \in [3]} \text{End}(P_i)$. If there exists a feasible graph Q which is disjoint from the heavy paths and $e_M(W, Q) \geq 33$, then $M[W \cup V(P_1 \cup P_2 \cup P_3)]$ is 2-feasible.*

Proof. Since $e_M(W, Q) \geq 33$, one of the three disjoint heavy paths, say P_1 , satisfies $e(\text{End}(P_1), Q) \geq 11$, and one of the other two paths, say P_2 , satisfies $e(\text{End}(P_2), Q) \geq 9$. By average at least one vertex of Q , say $u \in Q$ has $e(\text{End}(P_2), u) \geq 3$, then $G[V(P_2) \cup \{u\}]$ is a feasible graph Q' isomorphic to Q_2 . Since $e(\text{End}(P_1), Q) \geq 11$, except u , there is another vertex $v \in Q$ such that $e(\text{End}(P_1), v) \geq 3$ and $G[V(P_1) \cup \{v\}]$ is a feasible graph Q'' isomorphic to Q_2 and disjoint from Q' . $\square$

2.2 Proof of Theorem 1.1

Let M be a multigraph of order $4k$ with minimum degree $\delta(M) \geq 16k/3$. Suppose that Theorem 1.1 fails and M is an edge-maximal counterexample. Then M contains $k - 5$ feasible multigraphs, where $k \geq 6$. Let $\mathcal{Q}$ be a collection of $k - 5$ feasible multigraphs and $R = M - \mathcal{Q}$. Note that R has no feasible graph and $|R| = 20$. We denote by $\mathcal{T}$ a set of a maximum number of disjoint heavy path of length 2 in R , and $\mathcal{I}$ a set of maximum number of independent heavy edges in $R - \mathcal{T}$. Let $S = R - \mathcal{T} - \mathcal{I}$.

Choose $\mathcal{Q}$ in M such that

- (I) the size of $\mathcal{T}$ is as large as possible.
- (II) subject to (I) the size of $\mathcal{I}$ is as large as possible.

Claim 1. $|\mathcal{T}| \leq 2$.

Proof. Suppose $|\mathcal{T}| \geq 3$. Let P_1, P_2 and P_3 be three disjoint heavy paths of $\mathcal{T}$. Since R is feasible graph free, $e_M(\text{End}(P_i), v) \leq 2$, for every $v \in R - P_i$. Thus $d_R(\text{End}(P_i)) \leq 2 \times (20 - 3) + 8 = 42$ and $e_M(\text{End}(P_i), \mathcal{Q}) \geq 2 \times \frac{16k}{3} - 42$. Let $W = \{\text{End}(P_i) : i \in [3]\}$. Now we have

$$e_M(W, \mathcal{Q}) \geq 3 \times (2 \times \frac{16k}{3} - 42) = 32k - 126.$$

By average, there is some $Q \in \mathcal{Q}$ such that

$$e_M(W, Q) \geq \frac{32(k-5) + 160 - 126}{k-5} = 32 + \frac{34}{k-5} > 33.$$

Now Proposition 2 implies that $M[V(Q \cup P_1 \cup P_2 \cup P_3)]$ contains two disjoint feasible graphs, which together with $\mathcal{Q} - Q$ give $k - 4$ disjoint feasible graphs, a contradiction. $\square$

Claim 2. $|\mathcal{I}| \leq 2$.

Proof. Suppose $|\mathcal{I}| \geq 3$. Let $W \subset \mathcal{I}$ and $|W| = 3$. For each heavy edge $uv \in W$, since R has no feasible graph, it is easy to check that $e_M(\{u, v\}, \mathcal{T}) \leq 6|\mathcal{T}|$. By choice (I), $e_M(\{u, v\}, R - \mathcal{T} - \{u, v\}) \leq 2(|R| - 3|\mathcal{T}| - 2)$. In conclusion, $d_R(\{u, v\}) \leq 6|\mathcal{T}| + 2(|R| - 3|\mathcal{T}| - 2) + 4 = 2|R| = 40$. Hence, $d_R(W) \leq 3 \times 40 = 120$ and $e_M(W, \mathcal{Q}) \geq 6 \times \frac{16k}{3} - 120 = 32k - 120$. By average, there is some $Q \in \mathcal{Q}$ such that

$$e_M(W, Q) \geq \frac{32(k-5) + 160 - 120}{k-5} = 32 + \frac{40}{k-5} > 33.$$

Then Proposition 1 shows that $M[W \cup V(Q)]$ contains a feasible graph Q' and a heavy path P of length two. Let $\mathcal{Q}' = \mathcal{Q} \cup \{Q'\} - \{Q\}$ and $R' = M - \mathcal{Q}'$. Then $P \cup \mathcal{T}$ is a collection of disjoint heavy paths of length 2 in R' , which contradicts our choice (I). $\square$

Claim 3. *Subject to (I) and (II), there is a choice of $\mathcal{Q}$ in M such that $M[S]$ contains a $K_{2,4}^-$.*

Proof. By Claims 1 and 2, $|S| \geq 20 - 2 \times 3 - 2 \times 2 = 10$. Since R has no feasible graph, for every $v \in S$, $e_M(v, \mathcal{T}) \leq 4|\mathcal{T}|$. By choice (I), for every $v \in S$, $e_M(v, \mathcal{I}) \leq 2|\mathcal{I}|$. To summarize, for every $v \in S$, $e_M(v, R - S) \leq 4|\mathcal{T}| + 2|\mathcal{I}|$. If $M[S]$ contains a $K_{2,4}^-$, then we are done. So suppose that $M[S]$ contains no $K_{2,4}^-$ and thus no $K_{2,4}$. Moreover, S has no heavy edges by choice (II). Hence, for every two vertices $u, v \in S$, $d_S(u) + d_S(v) \leq 2 \times 3 + (|S| - 3 - 2) + 2 = |S| + 3$. Let $W \subset S$ and $|W| = 6$. Now we have $d_R(W) \leq 6 \times (4|\mathcal{T}| + 2|\mathcal{I}|) + 3 \times (|S| + 3) = 24|\mathcal{T}| + 12|\mathcal{I}| + 3|S| + 9 = 15|\mathcal{T}| + 6|\mathcal{I}| + 3|R| + 9$. By Claims 1 and 2, $d_R(W) \leq 15 \times 2 + 6 \times 2 + 3 \times 20 + 9 = 111$. Therefore, $e(W, \mathcal{Q}) \geq 32k - 111$. By average, there is some $Q \in \mathcal{Q}$ such that

$$e_M(W, Q) \geq \frac{32(k-5) + 160 - 111}{k-5} = 32 + \frac{49}{k-5} > 33.$$

By Lemmas 1 and 2, either (i) $M[W \cup V(Q)]$ contains a feasible graph Q' and a heavy edge e , which are disjoint or (ii) $Q \cong Q_2$ and $M[W \cup V(Q)]$ contains a feasible graph Q' and a $K_{2,4}^-$, which are disjoint. Let $\mathcal{Q}' = \mathcal{Q} \cup Q' - Q$ and $R' = M - \mathcal{Q}'$. For the case (i), R' is a choice contradicts (II). So we may assume that the case (ii) holds and then $\mathcal{Q}'$ is what we want. $\square$

Let $\mathcal{Q}$ be a choice satisfies Claim 3, and let K^* be a $K_{2,4}^-$ in $M[S]$. To count the degree of K^* in R , we need to divide the vertices of K^* into pairs: in K^* we can find two independent light edges, say e_1, e_2

and two remaining vertices u, v . Note that R contains no feasible graphs. Then it is not difficult to check that $e_M(u, \mathcal{T}) \leq 4|\mathcal{T}|$ and $e_M(v, \mathcal{T}) \leq 4|\mathcal{T}|$. Moreover, Fact 3 (ii) implies that $e_M(e_i, \mathcal{T}) \leq 7|\mathcal{T}|$ for each $i \in [2]$. That is, $e_M(K^*, \mathcal{T}) \leq 2 \times 7|\mathcal{T}| + 2 \times 4|\mathcal{T}| = 22|\mathcal{T}|$. By choice (I), there is no heavy path of length 2 in $R - \mathcal{T}$, so $e_M(w, \mathcal{I}) \leq 2|\mathcal{I}|$ for each $w \in V(K^*)$. Hence, $e_M(K^*, \mathcal{I}) \leq 6 \times 2|\mathcal{I}| = 12|\mathcal{I}|$. Finally, as there is no heavy edge in S , we have $d_S(w) \leq |S| - 1$ for each $w \in V(K^*)$. Thus $d_S(K^*) \leq 6 \times (|S| - 1)$. Therefore, $d_R(K^*) \leq 22|\mathcal{T}| + 12|\mathcal{I}| + 6(|S| - 1) = 4|\mathcal{T}| + 6|R| - 6 = 4|\mathcal{T}| + 114$. By Claims 1 and 2, $d_R(K^*) \leq 122$. The minimum degree condition now implies that $e(K^*, \mathcal{Q}) \geq 32k - 122$. By average, there is some $Q \in \mathcal{Q}$ such that

$$e_M(K^*, Q) \geq \frac{32(k - 5) + 160 - 122}{k - 5} = 32 + \frac{38}{k - 5} > 33.$$

By Lemma 1 or Lemma 3, $M[K^* \cup V(Q)]$ contains a feasible graph Q' and a heavy edge e , which are disjoint. Let $\mathcal{Q}' = \mathcal{Q} \cup Q' - Q$ and $R' = M - \mathcal{Q}'$. Then R' is a choice contradicts (II) and this completes the proof of Theorem 1.1.

3 Proof of Theorem 1.2

Given a multigraph $M = (V, E)$, we define two simple subgraphs H_M and G_M of M as follows: $V(H_M) = V(G_M) = V$ and $E(H_M) = \{xy : e_M(x, y) = 2\}$ and $E(G_M) = \{xy : e_M(x, y) \geq 1\}$. When it is clear from the context, we omit the subscript M . For any $U \subseteq V$, let $\bar{U} = V \setminus U$. Recall that $\mathcal{Q}^*(9)$ is the family of multigraphs shown in Figure 1 (a)-(d). We use the symbol $\mathcal{Q}^*(9)$ to denote any multigraph in $\mathcal{Q}^*(9)$. For $\alpha > 0$, a graph F on n vertices is α -splittable, if there is a partition A, B of $V(F)$ such that $|A|, |B| \geq n/3$ and $e_F(A, B) \leq \alpha n^2$.

Theorem 3.1. *For $1 > \alpha, \varepsilon > 0$, there exists $n_0 = n_0(\alpha, \varepsilon)$ such that every multigraph M on $n \geq n_0$ vertices contains a $\mathcal{Q}^*(9)$ -factor if $n \equiv 0 \pmod{4}$, $\delta(M) \geq (4/3 + \varepsilon)n$ and H_M is not α -splittable.*

Actually Theorem 3.1 yields Theorem 1.2. Let us first prove Theorem 3.1, then show how we get Theorem 1.2 from Theorem 3.1.

3.1 Preparations for Theorems 1.2 and 3.1

In this section, we will collect simple facts and definitions used in the following. Some of the facts and definitions also appear in [9]. Given a real number ε with $1 > \varepsilon > 0$, let $M = (V, E)$ be a multigraph on n vertices with minimum degree at least $(4/3 + \varepsilon)n$. The following holds obviously:

$$\delta(G) \geq (2/3 + \varepsilon/2)n \text{ and } \delta(H) \geq (1/3 + \varepsilon)n. \quad (1)$$

For a subgraph $F \subseteq M$ and an integer $k \geq 1$, define a vertex set $Q(F; k) := \{v \in V : e_M(v, F) \geq k\}$. When F is an edge e , we have the following inequality:

$$2(4/3 + \varepsilon)n \leq \sum_{v \in e} d_M(v) \leq 2|\overline{Q(e; 3)}| + 3|Q(e; 3)| + |Q(e; 4)|. \quad (2)$$

Note that $Q(e; 3) \supseteq Q(e; 4)$, solving the above inequality we obtain that

$$|Q(e; 3)| \geq (1/3 + \varepsilon)n. \quad (3)$$

When $F = T(5) \subseteq M$, we have $3(4/3 + \varepsilon)n \leq \sum_{v \in T(5)} d_M(v)$ and $\sum_{v \in T(5)} d_M(v) \leq 3|\overline{Q(T(5); 4)}| + 4|Q(T(5); 4)| + |Q(T(5); 5)| + |Q(T(5); 6)|$. Since $Q(T(5); 4) \supseteq Q(T(5); 5) \supseteq Q(T(5); 6)$, the following holds:

$$|Q(T(5); 4)| \geq (1/3 + \varepsilon)n. \quad (4)$$

For any vertex $u \in V$, we now define $F(u) := \{e \in E(H) : u \in Q(e; 3)\}$ and $F^*(u) := \{T(5) \in M : u \in Q(T(5); 4)\}$. Note that $M[V(e) \cup \{u\}]$ has a 5-triangle $T(5)$ for any $e \in F(u)$, and $M[V(T(5)) \cup \{u\}]$ contains a $Q^*(9)$ for each $T(5) \in F^*(u)$. Now we will estimate the lower bound of $F(u)$ and $F^*(u)$. It follows by (1) that $2|F(u)| \geq \sum_{v \in N_H(u)} |N_G(u) \cap N_H(v)| \geq (1 + 3\varepsilon)\varepsilon n^2/2$. So $|F(u)| \geq \varepsilon n^2/4$. Similarly, $3|F^*(u)| \geq \sum_{e \in F(u)} |Q(e; 3) \cap N_G(u)| \geq (\varepsilon n^2/4)(3\varepsilon n/2) = 3\varepsilon^2 n^3/8$ implies

$$|F^*(u)| \geq \varepsilon^2 n^3/8. \quad (5)$$

Definition 3.1. For disjoint sets $X, Y \subseteq V$, we say that Y absorbs X if both $M[Y]$ and $M[X \cup Y]$ contain a $Q^*(9)$ -factor. Let $Z := (z_1, \dots, z_{80}) \in V^{80}$ and $V(Z) := \{z_1, \dots, z_{80}\}$. For $X \in \binom{V}{4}$, if $V(Z)$ absorbs X , then we call Z an X -absorber. Denote the set of all X -absorbers by $\mathcal{A}_X$.

Definition 3.2. For a positive integer r , the tuple $(z_1, \dots, z_{4r-1})$ is an r -chain if

- (i) $z_1, \dots, z_{4r-1}$ are distinct vertices in V ;
- (ii) Let $T_i = M[\{z_{4i-3}, z_{4i-2}, z_{4i-1}\}]$. Every T_i contains a 5-triangle for all $i \in [r]$;
- (iii) $z_{4i} \in Q(T_i; 4) \cap Q(T_{i+1}; 4)$ for all $i \in [r-1]$.

The above definition indicates that an r -chain is an ordered tuple, so different r -chains may have same vertex set, but different order. Let $u, v \in V$. If $u, v \notin \{z_1, \dots, z_{4r-1}\}$, $u \in Q(T_i; 4)$ and $v \in Q(T_j; 4)$ for $i, j \in [r]$, we say that the r -chain joins u to v . Note that i may equal j , and if an r -chain joins u to v , then it also joins v to u . Let $0 < \sigma < \varepsilon/8$. If there exist at least $(\sigma n)^{4r-1}$ different r -chains that join u to v , then we say that u is r -joined to v .

Lemma 6. For any integers i, j with $1 \leq i < j \leq 5$, if u is i -joined to v , then u is also j -joined to v .

Proof. It suffices to prove that when $1 \leq i < 5$, if u is i -joined to v , then u is also $(i+1)$ -joined to v . Assume that $(z_1, \dots, z_{4i-1})$ is an i -chain that joins u to v . By (4), $|Q(T(5); 4)| \geq (1/3 + \varepsilon)n$. When n is sufficiently large, $|Q(T(5); 4) \setminus \{z_1, \dots, z_{4i-1}\}| \geq n/3$. This means that to extend the i -chain $(z_1, \dots, z_{4i-1})$ joining u to v to an $(i+1)$ -chain $(z_1, \dots, z_{4i-1}, z_{4i}, z_{4i+1}, z_{4i+2}, z_{4i+3})$, there are at least $n/3$ choices for z_{4i} . For each fixed z_{4i} , inequality (5) implies that there are at least $\varepsilon^2 n^3/8$ different $T(5)$ s in M such that $z_{4i} \in Q(T(5); 4)$, again when n is large, at least $\varepsilon^2 n^3/16$ of them are disjoint from the previous fixed vertices. Now we have shown that for a fixed i -chain joining u to v , there are at least $(n/3)(\varepsilon^2 n^3/16) = \varepsilon^2 n^4/48$ ways to extend it into an $(i+1)$ -chain joining u to v . Since u is i -joined to v , there exist $(\sigma n)^{4i-1}$ different i -chains that join u to v . Therefore, by extending these i -chains, we can get at least $(\sigma n)^{4i-1}(\varepsilon^2 n^4/48) \geq (\sigma n)^{4(i+1)-1}$ different $(i+1)$ -chains joining u to v . $\square$

For $u \in V$, let $L_k(u) := \{v \in V : u \text{ is } k\text{-joined to } v\}$. Lemme 6 implies $L_i \subseteq L_{i+1}$ for $i \in [4]$.

Lemma 7. For every $v \in V$, we have $v \in L_1(v)$ and $|L_1(v)| \geq (1/3 + \varepsilon/3)n$.

Proof. It follows from (5) that $|F^*(v)| \geq \varepsilon^2 n^3/8 \geq (\sigma n)^3$, so the vertex v belongs to $L_1(v)$. Let $t := \sum_{T(5) \in F^*(v)} |Q(T(5); 4)|$. According to (4), $t \geq |F^*(v)|(1/3 + \varepsilon)n$. On the other hand, by double counting, $t = \sum_{u \in V} |F^*(u) \cap F^*(v)|$. If $u \notin L_1(v)$, then $|F^*(u) \cap F^*(v)| \leq (\sigma n)^3 < \varepsilon |F^*(v)|$. Therefore,

$$t \leq |F^*(v)||L_1(v)| + \varepsilon |F^*(v)||\overline{L_1(v)}| = \varepsilon |F^*(v)|n + (1 - \varepsilon)|F^*(v)||L_1(v)|.$$

So we get the following inequality

$$(1/3 + \varepsilon)n \leq \varepsilon n + (1 - \varepsilon)|L_1(v)|,$$

solving it gives that $|L_1(v)| \geq n/(3 - 3\varepsilon) \geq (1/3 + \varepsilon/3)n$. $\square$

Lemma 8. Let $u, v \in V$ be two distinct vertices and let $i, j \in [2]$. If $|L_i(u) \cap L_j(v)| \geq 2\sigma n$, then $v \in L_{i+j}(u)$.

Proof. For each $w \in L_i(u) \cap L_j(v)$, since $w \in L_i(u)$, there are $(\sigma n)^{4i-1}$ ways to choose an i -chain that joins u to w . Since $w \in L_j(v)$, there are $(\sigma n)^{4j-1}/2$ ways to choose a j -chain that joins w to v avoiding the previously picked vertices. Together they form an $(i + j)$ -chain joining u to v . Thus there are at least

$$(2\sigma n)(\sigma n)^{4i-1}(\sigma n)^{4j-1}/2 = (\sigma n)^{4(i+j)-1}$$

$(i + j)$ -chains that join u to v . Thus $v \in L_{i+j}(u)$ by the definition. $\square$

The proof of Theorem 3.1 is probabilistic, so the following inequalities are useful.

Lemma 9. (Chernoff bound) Let X be a random variable with binomial distribution and $E(X)$ be its expectation. Let e be the base of the natural log. Then for $\lambda > 0$,

$$P(X \geq E(X) + \lambda) \leq e^{-\lambda^2/(2E(X) + \lambda/3)},$$

$$P(X \leq E(X) - \lambda) \leq e^{-\lambda^2/(2E(X))}.$$

Lemma 10. (Markov inequality) Let X be a random variable and $E(X)$ be its expectation. Then for $\lambda > 0$,

$$P(X \geq \lambda) \leq E(X)/\lambda.$$

3.2 Proof of Theorem 3.1

Let ε be a positive real number and let $M = (V, E)$ be a multigraph on n vertices with $\delta(M) \geq (4/3 + \varepsilon)n$ and n sufficiently large. Set $0 < \sigma < \min\{\varepsilon/8, \sqrt[3]{\alpha/3^6}\}$ and $\tau := \sigma^{78}/2^7$. In this subsection, we are going to prove Theorem 3.1 by using the probabilistic absorbing method as in [9]. Our strategy is to first show that there exists a small set $\mathcal{A}$ of absorber such that for any $X \in \binom{V}{4}$, there is an X -absorber in $\mathcal{A}$, and then to prove that there is a set Y of size four in $V \setminus V(\mathcal{A})$ such that $M[V \setminus (Y \cup V(\mathcal{A}))]$ contains a $\mathcal{Q}^*(9)$ -factor. Since there is a Y -absorber in $\mathcal{A}$, this implies M has a $\mathcal{Q}^*(9)$ -factor.

Claim 1. Any vertex v of V satisfies $L_5(v) = V$.

Proof. Suppose there exists $u \notin L_5(v)$. We will first show that $L_2(u) \cup L_2(v) = V$. To the contrary, assume $w \notin L_2(u) \cup L_2(v)$. Then Lemmas 6 and 8 imply that $|L_1(u) \cap L_1(w)| < 2\sigma n$, $|L_1(v) \cap L_1(w)| < 2\sigma n$ and $|L_1(u) \cap L_1(v)| < 2\sigma n$. By Lemma 7, $|L_1(u) \cup L_1(v) \cup L_1(w)| \geq 3(1/3 + \varepsilon/3)n - 6\sigma n > n$, a contradiction.

Set $A := L_2(u)$ and $B := V \setminus A$. The previous discussion shows that $\{A, B\}$ forms a partition of V . As we already know $L_1(u) \subseteq L_2(u)$ and $L_1(v) \subseteq L_2(v)$, Lemma 7 gives that $|L_2(u)|, |L_2(v)| \geq (1/3 + \varepsilon/3)n$. Moreover, since $u \notin L_5(v)$, Lemmas 6 and 8 imply $|L_2(u) \cap L_2(v)| < 2\sigma n$. Therefore, $n/3 < |A|, |B| < 2n/3$. We define a 4-tuple (x_1, x_2, x_3, x_4) to be *good* if $x_1 \in A \cap Q(M[\{x_2, x_3, x_4\}]; 4)$, $x_4 \in B \cap Q(M[\{x_1, x_2, x_3\}]; 4)$, and both $M[\{x_1, x_2, x_3\}]$ and $M[\{x_2, x_3, x_4\}]$ contain $T(5)$. The following proof indicates that there are many good 4-tuples in M .

Let $xy \in E_H(A, B)$ be a heavy edge in M between A and B . If $|Q(xy; 4)| \geq \sigma n$, then there are at least σn different $T(6)$ s containing xy . Let xyz be a $T(6)$ using xy . Then

$$3(4/3 + \varepsilon)n \leq d_M(x) + d_M(y) + d_M(z) \leq 6|Q(xyz; 5)| + 4|\overline{Q(xyz; 5)}|,$$

which gives that $|Q(xyz; 5)| \geq 3\varepsilon n/2$. For $t \in Q(xyz; 5)$, if $t \in A$, then (t, x, z, y) is a good 4-tuple; if $t \in B$, then (x, y, z, t) is a good 4-tuple. So in this case the number of good 4-tuples containing xy is at least $(\sigma n)(3\varepsilon n/2) = 3\varepsilon \sigma n^2/2 > (\sigma n)^2$. Now we assume that $|Q(xy; 4)| < \sigma n$. It can be inferred from (2) that $|Q(xy; 3)| > (2/3 + \varepsilon)n$. Without loss of generality, we may assume $|Q(xy; 3) \cap A| > (1/3 + \varepsilon/2)n$. As $n/3 < |A|, |B| < 2n/3$, it is easy to see that $|Q(xy; 3) \cap B| > \varepsilon n$. Then it follows from (1) that for each $z \in Q(xy; 3) \cap B$, we have $|N_M(z) \cap (Q(xy; 3) \cap A)| \geq (2/3 + \varepsilon/2)n + (1/3 + \varepsilon/2)n - n = \varepsilon n$. This means that for each $z \in Q(xy; 3) \cap B$, there are at least εn distinct $t \in Q(xy; 3) \cap A$ such that zt forms an edge in M . It is then not hard to check that (t, x, y, z) is a good 4-tuple, and in this case the number of good 4-tuples containing xy is at least $(\varepsilon n)^2 > (\sigma n)^2$. Now we have shown that the number of good 4-tuples containing an edge $e \in E_H(A, B)$ is at least $(\sigma n)^2$. Since H is not α -splittable, $e_H(A, B) \geq \alpha n^2$. In addition, each good 4-tuple contains no more than 6 edges in $E_H(A, B)$, so the number of good 4-tuples in M is at least $(\alpha n^2)(\sigma n)^2/6$.

We say that a 5-tuple $(x_1, x_2, x_3, x_4, x_5)$ is *connecting* if $x_1 \in A$, $x_5 \in B$, and x_2, x_3, x_4 forms a 1-chain that joins x_1 to x_5 . Actually connecting 5-tuples is easy to construct from good 4-tuples. Given a good 4-tuple (x_1, x_2, x_3, x_4) , let us consider a vertex $x_5 \in Q(M[\{x_1, x_2, x_3, x_4\}]; 6)$: If $x_5 \in A$, then the 5-tuple $(x_5, x_1, x_2, x_3, x_4)$ is connecting; otherwise $(x_1, x_2, x_3, x_4, x_5)$ is connecting. Since

$$\begin{aligned} &|Q(M[\{x_1, x_2, x_3, x_4\}]; 8)| + |Q(M[\{x_1, x_2, x_3, x_4\}]; 7)| + 6|Q(M[\{x_1, x_2, x_3, x_4\}]; 6)| \\ &+ 5|\overline{Q(M[\{x_1, x_2, x_3, x_4\}]; 6)}| \geq \sum_{i \in [4]} d_M(x_i) \geq 4(4/3 + \varepsilon)n, \end{aligned}$$

and $Q(M[\{x_1, x_2, x_3, x_4\}]; 8) \subseteq Q(M[\{x_1, x_2, x_3, x_4\}]; 7) \subseteq Q(M[\{x_1, x_2, x_3, x_4\}]; 6)$, after few easy computation, we see that $|Q(M[\{x_1, x_2, x_3, x_4\}]; 6)| \geq (1/9 + 4\varepsilon/3)n$. So each good 4-tuple can be contained in at least $(1/9 + 4\varepsilon/3)n$ connecting 5-tuples. Moreover, every 5-tuple contains no more than five 4-tuples, therefore M contains at least $((\alpha n^2)(\sigma n)^2/6)(1/9 + 4\varepsilon/3)n/5 > 4(\sigma n)^5$ connecting 5-tuples.

Now it comes to the final step, we will show that each connecting 5-tuple can be extended to

a 5-chain that joins u to v in many ways. Let $(x_1, x_2, x_3, x_4, x_5)$ be a connecting 5-tuples. Since $x_1 \in A = L_2(u)$, there exist $(\sigma n)^7/2$ different 2-chains joining u to x_1 by avoiding vertices already appeared. Similarly, as $x_5 \in B \subseteq L_2(v)$, there exist $(\sigma n)^7/2$ different 2-chains joining v to x_5 by avoiding previous vertices. Therefore, there are at least $4(\sigma n)^5((\sigma n)^7/2)^2 = (\sigma n)^{19}$ different 5-chains joining v to v , so $u \in L_5(v)$. $\square$

Let $X := \{x_1, x_2, x_3, x_4\}$ be a subset of V , $Y := (v_1, \dots, v_{80}) \in V^{80}$ and $m(i) = 20(i-1)$. Then it is easy to check that $Y \in \mathcal{A}_X$ if the following hold for $i \in [4]$:

- (a) $v_1, \dots, v_{80}$ are all distinct vertices in V ;
- (b) $M[\{v_{m(1)+1}, v_{m(2)+1}, v_{m(3)+1}, v_{m(4)+1}\}]$ contains a $Q^*(9)$;
- (c) $(v_{m(i)+2}, \dots, v_{m(i)+20})$ is a 5-chain that joins $v_{m(i)+1}$ to x_i for $i \in [4]$.

The next claim shows that any 4-set can be absorbed by a positive fractional of all 80-tuples.

Claim 2. For any $X \in \binom{V}{4}$, $|\mathcal{A}_X| \geq 4\tau n^{80}$.

Proof. Let $X := \{x_1, x_2, x_3, x_4\}$ and $m(i) := 20(i-1)$ for $i \in [4]$. To estimate the lower bound of $|\mathcal{A}_X|$, we count the number of 80-tuples $Y := (v_1, \dots, v_{80})$ that satisfies (a), (b) and (c). By (5), we obtain that $|F^*(v)| \geq \varepsilon^2 n^3/8$ for $v \in V \setminus X$. As n is very large, by avoiding the vertices in X , at least $\varepsilon^2 n^3/16$ of them can form a $Q^*(9)$ with v . So the number of choices of $(v_{m(1)+1}, v_{m(2)+1}, v_{m(3)+1}, v_{m(4)+1})$ is at least $(n-4)\varepsilon^2 n^3/64 > \varepsilon^2 n^4/128$. According to Claim 1, $L_5(v_{m(i)+1}) = V$ for $i \in [4]$, and $x_i \in L_5(v_{m(i)+1})$. That is to say, $v_{m(i)+1}$ is 5-joined to x_i , and there are at least $(\sigma n)^{19}$ different 5-chains joining $v_{m(i)+1}$ to x_i . Again as n is large, at least half of them are disjoint from the previous fixed vertices. So the number of choices for $(v_{m(i)+2}, \dots, v_{m(i)+20})$ is at least $(\sigma n)^{19}/2$. Therefore,

$$|\mathcal{A}_X| \geq (\varepsilon^2 n^4/128)((\sigma n)^{19}/2)^4 = \varepsilon^2 \sigma^{76} n^{80}/2^{11} \geq (8\sigma)^2 \sigma^{76} n^{80}/2^{11} \geq 4\tau n^{80}.$$

This proves the claim. $\square$

Next by using Chernoff bound and Markov's inequality we obtain a union bound to show that with positive probability there is a small size set of disjoint absorbers absorbing every 4-set.

Claim 3. There exists a set $\mathcal{A}$ of disjoint absorbers such that for every $X \in \binom{V}{4}$ the intersection of $\mathcal{A}_X$ and $\mathcal{A}$ is at least 4 and $|\mathcal{A}| \leq \varepsilon n/160$.

Proof. Let $\mathcal{F}$ be a set of 80-tuples obtained by choosing each 80-tuple $Y \in V^{80}$ randomly and independently with probability ρn^{-79} , where $\rho := \tau/6400$. Define $\varepsilon' := \varepsilon/160$, then $\rho < \varepsilon'$. Note that $E(|\mathcal{F}|) = \rho n$ and $E(|\mathcal{F}|) \geq E(|\mathcal{F} \cap \mathcal{A}_X|) \geq 4\tau \rho n$ for each $X \in \binom{V}{4}$ according to Claim 2. Denote the set of intersecting pairs in V^{80} by $\mathcal{O}$, that is

$$\mathcal{O} := \{(x_1, \dots, x_{80}), (y_1, \dots, y_{80})\} \in \binom{V^{80}}{2} : x_i = y_j \text{ for some } i, j \in [80]\}.$$

Then $|\mathcal{O}| \leq n \cdot 80 \cdot 80 \cdot n^{158}/2 = 3200n^{159}$. For a pair $O \in \mathcal{O}$, the probability that it is contained in $\mathcal{F}$ is $P(O \subseteq \mathcal{F}) = (\rho n^{-79})^2$. Let $\mathcal{O}_{\mathcal{F}} := \{O \in \mathcal{O} : O \subseteq \mathcal{F}\}$. Then $E(|\mathcal{O}_{\mathcal{F}}|) \leq 3200n^{159}(\rho n^{-79})^2 = 3200\rho^2 n \leq \tau \rho n/2$.

Next we compute the probabilities of three 'bad' events: By Lemma 9,

$$P(|\mathcal{F}| \geq \varepsilon' n) = P(|\mathcal{F}| \geq E(|\mathcal{F}|) + (\varepsilon' - \rho)n) \leq e^{-3(\varepsilon' - \rho)^2 n / (5\rho + \varepsilon')},$$

and for $X \in \binom{V}{4}$,

$$P(|\mathcal{F} \cap \mathcal{A}_X| \leq 3\tau\rho n) = P(|\mathcal{F} \cap \mathcal{A}_X| \leq E(|\mathcal{F} \cap \mathcal{A}_X|) - \tau\rho n) \leq e^{-(\tau\rho)^2 n/(2\rho)};$$

By Lemma 10,

$$P(|\mathcal{O}_{\mathcal{F}}| \geq \tau\rho n) \leq 1/2.$$

Therefore, we may get the following union bound when n is large enough

$$P(|\mathcal{F}| \geq \varepsilon' n) + \sum_{X \in \binom{V}{4}} P(|\mathcal{F} \cap \mathcal{A}_X| \leq 3\tau\rho n) + P(|\mathcal{O}_{\mathcal{F}}| \geq \tau\rho n) < 1.$$

The above union bound shows that there exists $\mathcal{F} \in V^{80}$ with $|\mathcal{F}| < \varepsilon' n$, $|\mathcal{F} \cap \mathcal{A}_X| > 3\tau\rho n$ for every $X \in \binom{V}{4}$, and $|\mathcal{O}_{\mathcal{F}}| < \tau\rho n$. Let $\mathcal{A}'$ be obtained from $\mathcal{F}$ by deleting all the elements in $\mathcal{O}_{\mathcal{F}}$. Obviously, all the 80-tuples in $\mathcal{A}'$ are vertex disjoint and $|\mathcal{A}'| < \varepsilon' n$ and $|\mathcal{A}' \cap \mathcal{A}_X| > 3\tau\rho n - 2\tau\rho n = \tau\rho n \geq 4$ when n is sufficiently large, for every $X \in \binom{V}{4}$. Let $\mathcal{A}$ be obtained from $\mathcal{A}'$ by deleting all the elements Which is not an absorber. Now $\tau\rho n < |\mathcal{A}' \cap \mathcal{A}_X| \leq |\mathcal{A}| \leq |\mathcal{A}'|$, so $\mathcal{A}$ is the desired set. $\square$

Let $\mathcal{A}$ be the set guaranteed by Claim 3 and $M' := M \setminus V(\mathcal{A})$. Then $|V(\mathcal{A})| \leq \varepsilon n/2$ and $\delta(M') \geq (4/3 + \varepsilon/2)n$. It follows from Theorem 1.1 that there exists a 16-set $X \subseteq V(M')$ and a $\mathcal{Q}^*(9)$ -factor of $M[V(M') \setminus X]$. Divide X into disjoint 4-sets $X = X_1 \cup X_2 \cup X_3 \cup X_4$. Since $|\mathcal{A} \cap \mathcal{A}_{X_i}| \geq 4$ for $i \in [4]$, we get that $M[V(\mathcal{A}) \cup X]$ has a $\mathcal{Q}^*(9)$ -factor. This completes the proof of Theorem 3.1. $\square$

3.3 Use Theorem 3.1 to prove Theorem 1.2

Suppose that $D = (V, E)$ is a digraph on n vertices with minimum semi-degree $\delta^0(D) \geq (2/3 + \varepsilon)n$. Let M represent the corresponding standard multigraph. Then the minimum semi-degree of D implies $\delta(M) \geq (4/3 + 2\varepsilon)n$. Using this and (1), we get $\delta(H) \geq (1/3 + 2\varepsilon)n$, where $H := H_M$. Set $\alpha := \varepsilon/6$. If H is not α^2 -splittable, applying Theorem 3.1 with parameter α^2 , we get that M contains a $\mathcal{Q}^*(9)$ factor. As we discussed before, this gives a 4-cycle factor in the digraph D . So it suffices to consider the case that H is α^2 -splittable. Then there exists a partition $\{V_1, V_2\}$ of $V(H) = V$ such that $|V_1|, |V_2| \geq n/3$ and $e_H(V_1, V_2) \leq \alpha^2 n^2$. Define $Z := \{v \in V_i : |N_H(v) \cap V_{3-i}| > 2\alpha n\}$ be the set of vertices in H that has more than $2\alpha n$ neighbors in its opposite part. Obviously, $|Z| \leq \alpha n$ since otherwise $e_H(V_1, V_2) > \alpha^2 n^2$.

Now we claim that the vertices in Z can be covered by using disjoint multigraphs in $\mathcal{Q}^*(9)$. As mentioned before, these correspond to disjoint 4-cycles in D . By the definition of Z and the fact $|Z| \leq \alpha n$, there is a matching K' in H that saturates Z , and it corresponds a matching K in M saturating Z such that each edge in K is a heavy edge of M . For every $e \in K$, (3) implies $|Q(e; 3)| \geq (1/3 + 2\varepsilon)n$. So for each $e \in K$, we can find a vertex v_e such that $L := M[V(K) \cup \{v_e : e \in K\}]$ forms a collection of disjoint $T(5)$ s in M . Further, for each $T(5) \in L$, (4) implies that $|Q(T(5); 4)| \geq (1/3 + 2\varepsilon)n$. Therefore, there exists a vertex $v_{T(5)}$ such that $X := M[V(L) \cup \{v_{T(5)} : T(5) \in L\}]$ forms a collection of disjoint multigraphs in $\mathcal{Q}^*(9)$. This proves our claim.

Let $V' := V \setminus V(X)$, $A' := V_1 \setminus V(X)$ and $B' := V_2 \setminus V(X)$. Since $|Z| \leq \alpha n$, we have $|X| \leq 4\alpha n$. Note that by the definition of Z , $e_H(v, B') \leq 2\alpha n$ for each $v \in A'$, and $e_H(v, A') \leq 2\alpha n$ for each

$v \in B'$. Using the degree conditions of D and H , we obtain the following:

$$\delta^0(D[V']) \geq (2/3 + \varepsilon)n - 4\alpha n = (2/3 + 2\alpha)n, \quad (6)$$

$$\delta(H[V']) \geq (1/3 + 2\varepsilon)n - 4\alpha n = (1/3 + 8\alpha)n, \quad (7)$$

$$\delta(H[A']), \delta(H[B']) \geq (1/3 + 8\alpha)n - 2\alpha n = (1/3 + 6\alpha)n. \quad (8)$$

It is worthwhile mentioning that (8) implies $n/3 < |A'|, |B'| < 2n/3$, $|A'| > |B'|/2$ and $|B'| > |A'|/2$. If $|A'| = |B'| \equiv 0 \pmod{4}$, then let $A := A', B := B'$. Otherwise, since $|A'| + |B'| \equiv 0 \pmod{4}$, there are two cases to consider by symmetry of A' and B' , that is $|A'| \equiv 1 \pmod{4}$ and $|B'| \equiv 3 \pmod{4}$ or $|A'| = |B'| \equiv 2 \pmod{4}$. In the former case, we want a 4-cycle C_4 in $D[V']$ such that $|V(C_4) \cap A'| = 1$ and $|V(C_4) \cap B'| = 3$; in the later case, we want a C_4 in $D[V']$ such that $|V(C_4) \cap A'| = |V(C_4) \cap B'| = 2$. Then let $A := A' \setminus V(C_4)$ and $B := B' \setminus V(C_4)$, obviously $|A| = |B| \equiv 0 \pmod{4}$.

First we prove the existence of a 4-cycle C_4 in $D[V']$ with $|V(C_4) \cap A'| = 1$ and $|V(C_4) \cap B'| = 3$. Let $u \in A'$ be an arbitrary vertex. By (6) and $|A'| < 2n/3$, there exists an out-neighbour v of u in B' . By (8), v has an out-neighbour w in B' . Using (6) and (8), we can deduce that $|N^-(u) \cap (N^+(w) \cap B')| \geq (2/3 + 2\alpha)n + (1/3 + 6\alpha)n - n = 8\alpha n > 0$. So there is a vertex $t \in N^-(u) \cap N^+(w) \cap B'$ such that $uvwtu$ is a 4-cycle. Very similarly, one may prove that the existence of a 4-cycle C_4 in $D[V']$ with $|V(C_4) \cap A'| = |V(C_4) \cap B'| = 2$. The only difference is when we fix u, v, w , we need to find $t \in (N^-(u) \cap A') \cap N^+(w)$.

Finally, we will prove that both $D[A]$ and $D[B]$ contain a 4-cycle factor, which finishes the proof of Theorem 1.2. Remind that $e_H(v, B) \leq 2\alpha n$ for each $v \in A$, and $e_H(v, A) \leq 2\alpha n$ for each $v \in B$. Therefore,

$$\begin{aligned} \delta(D[A]) &\geq 2\delta^0(D[V']) - |B'| - 2\alpha n - 6 \\ &\geq (4/3 + 4\alpha)n - |B'| - 2\alpha n - 6 \\ &= (4/3 + 2\alpha)(n - |B'|) + (1/3 + 2\alpha)|B'| - 6 \\ &\geq (4/3 + 2\alpha)|A'| + (1/6 + \alpha)|A'| - 6 \\ &> 3|A|/2, \end{aligned}$$

and

$$\begin{aligned} \delta(D[B]) &\geq 2\delta^0(D[V']) - |A'| - 2\alpha n - 6 \\ &\geq (4/3 + 4\alpha)n - |A'| - 2\alpha n - 6 \\ &= (4/3 + 2\alpha)(n - |A'|) + (1/3 + 2\alpha)|A'| - 6 \\ &\geq (4/3 + 2\alpha)|B'| + (1/6 + \alpha)|B'| - 6 \\ &> 3|B|/2. \end{aligned}$$

Then both $D[A]$ and $D[B]$ contain a 4-cycle factor by the result of Gao, Zou and Ma [13], which says that every digraph of order n with minimum degree $\delta(D) > 3n/2$ contains a 4-cycle factor. This finishes the proof of Theorem 1.2. $\square$

4 Remark

The degree condition in Theorem 1.1 is best possible up to an additive constant due to the following example. Let A_1, A_2 and A_3 be three disjoint multigraphs such that each pair of vertices in A_i ($i \in [3]$) are joined by a heavy edge. Construct the multigraph M by adding all possible light edges between any two distinct multigraphs. Let $4k/3 - 1, 4k/3 - 1$ and $4k/3 + 2$ be the orders of A_1, A_2 and A_3 , respectively. It is easy to check that the multigraph M has minimum degree $16k/3 - 3$, however, it is not $(k - 1)$ -feasible.

With a more technical discussion, we can prove that every multigraph M of order $4k$ with minimum degree $\delta(M) \geq 16k/3 - 2$ is $(k - 1)$ -feasible. Another natural question is whether the same condition can guarantee the existence of $k - 1$ disjoint copies of Q_1 or $k - 1$ disjoint copies of Q_2 ? The following two examples imply that the answer is also no. Consider two disjoint complete graphs B_1 and B_2 of orders $4k/3$ and $8k/3$, respectively. Let M be the multigraph which obtained by attaching each vertex in $V(B_1)$ to each vertex in $V(B_2)$ by a heavy edge. Clearly, the minimum degree of M is $16k/3 - 1$, however, M has at most $2k/3$ disjoint copies of Q_1 . Next we construct a multigraph with only one Q_2 though its minimum degree is $16k/3$. Suppose that I_1, I_2, I_3 and I_4 are four disjoint independent sets with sizes $1, 4k/3, 4k/3$ and $4k/3 - 1$, respectively. Let M be the multigraph with vertex set $V(M) = I_1 \cup I_2 \cup I_3 \cup I_4$ and all possible heavy edges such that the end-vertices of each heavy edge belongs to different independent sets.

To conclude the paper, we pose a conjecture on all possible cycle factors in a digraph, which is a directed version of the Aigner-Brandt Theorem [1]. This is asymptotically correct for the cases $n_i = 3$ ($i \in [k]$) and $n_i = 4$ ($i \in [k]$) due to a result of Czygrinow, Kierstead and Molla [9] and Theorem 1.2, respectively.

Conjecture 1. *Every digraph of order n with minimum semi-degree $2n/3$ contains all possible cycle factors. That is, for any positive integer partition $n = n_1 + \dots + n_k$ with $n_i \geq 3$ for each i , digraph D has k disjoint cycles of length $n_1, \dots, n_k$, respectively.*

The following example shows that the minimum semi-degree in Conjecture 1 would be best possible if true. Let X and Y be two disjoint vertex sets of size $2k - 1$ and $k + 1$, respectively. Suppose D is a digraph with vertex set $X \cup Y$ and arc set $\{xy : x \in X, y \in X \cup Y\} \cup \{yx : x \in X, y \in Y\}$. Clearly, $\delta^0(D) = 2k - 1 = 2n/3 - 1$, but it contains no k disjoint 3-cycles.

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